PREDICTIVE ANALYTICS AND DATA VISUALIZATION · 25MCAAI302 · ACTIVITY 2

When does Montesinho burn — and what can a forest agency do about it?

Data Storytelling Through Visualization · UCI Forest Fires dataset, Portugal

513
observations

6,613 ha
burned

10
visualizations

1
interactive dashboard

Anoon M R · 25MCAR0179 · MCA AIML, Sem III · Faculty: Prof. Rahul Pawar

THE DATA

513 days of park weather — when is fire danger highest?

513

clean observations
(517 raw, 4 duplicates removed)

52%

of observations had a fire

6,613 ha

total area burned

1,091 ha

largest single fire

13 attributes

- **Where:** X, Y position on a 9 × 9 park grid
- **When:** month and day of week
- **Weather:** temperature, humidity, wind, rain
- **Fire Weather Index codes:** FFMC, DMC, DC, ISI
- **Outcome:** burned area in hectares

Source

UCI Machine Learning Repository — Forest Fires

Cortez & Morais (2007), Montesinho Natural Park, north-east Portugal

Open dataset · climate / environment domain

DATA PREPARATION

Clean, check, and keep the extreme fires — they are the story

1

Quality checks

0 missing values · all range checks pass · 4 exact duplicates removed

2

Outliers kept

Rainy days, windy days and fires up to 1,091 ha are real events, not errors

3

Skew handled

Median fire only 6.36 ha → plot log(1 + area)

4

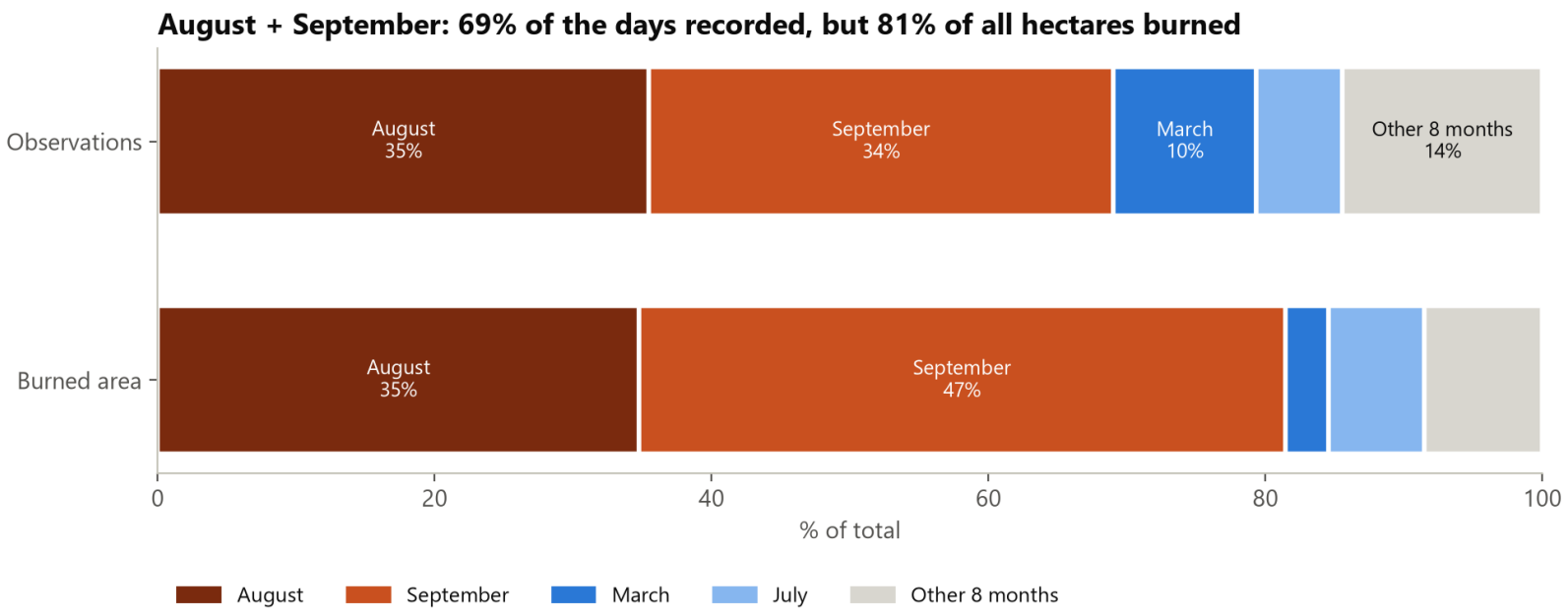
New features

season · fire-size class · fuel-dryness band · temperature band · peak period · weekday/weekend

Fire-size classes used in every chart

No fire	n = 244
Small < 1 ha	n = 27
Medium 1–10 ha	n = 148
Large > 10 ha	n = 94

August + September: 69% of the days, 81% of the hectares burned



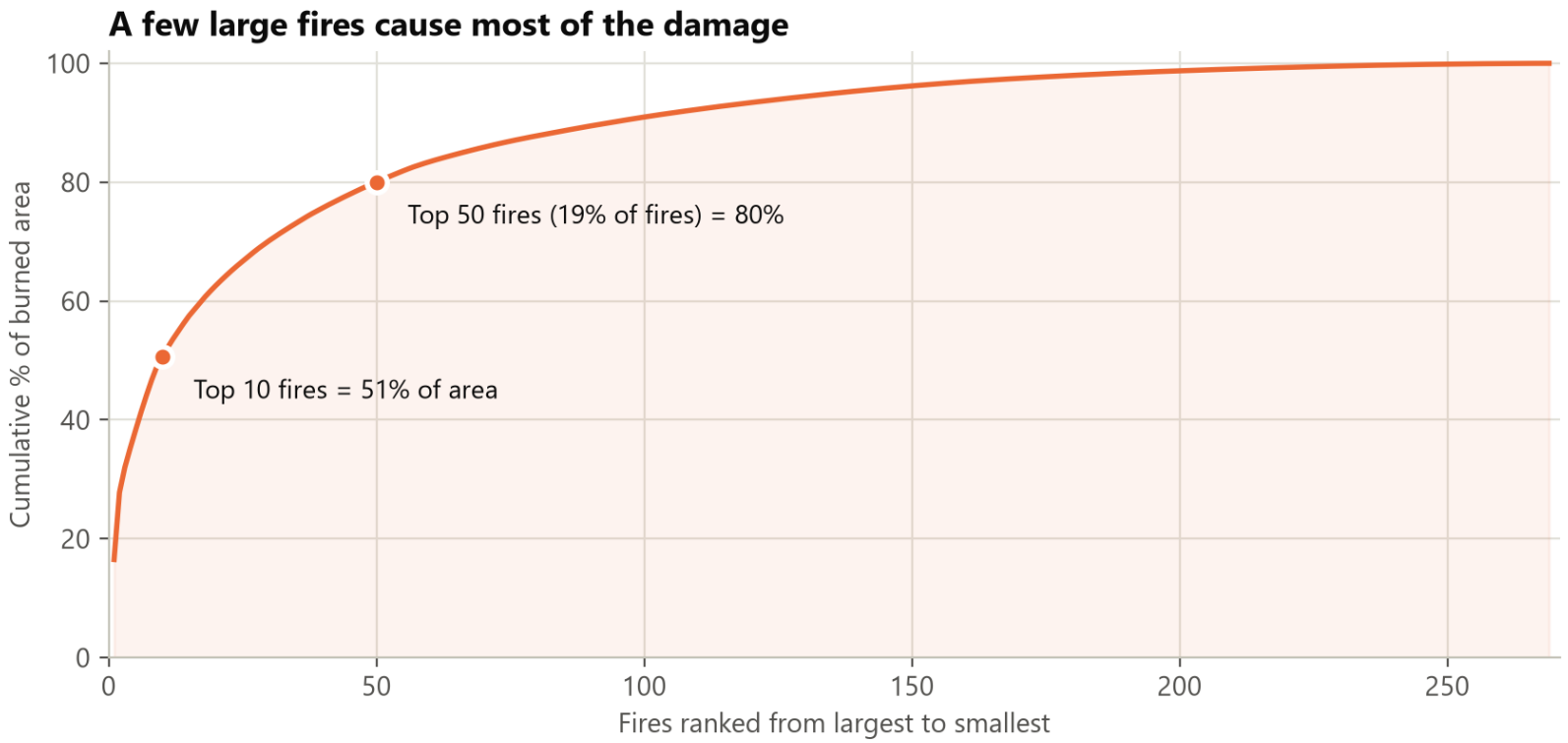
81%

of burned hectares fall in Aug–Sep

What it means

- Two months decide the fire year
- September fires are larger than their share of days
- March: 10% of days, only 3% of damage

A handful of large fires cause most of the damage



51%

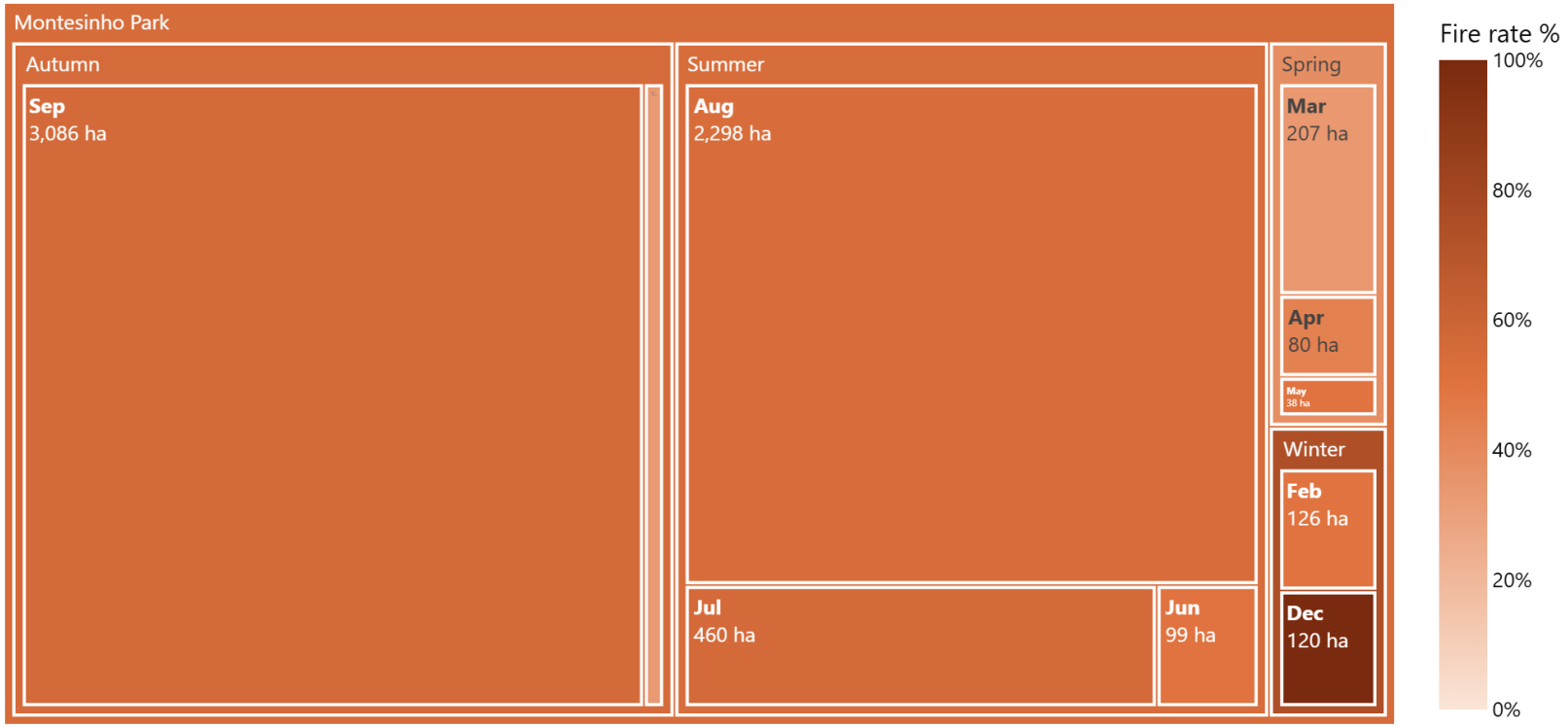
of all burned area comes from the 10 largest fires

What it means

- Top 50 fires (19%) = 80% of area
- Median fire is just 6.36 ha
- Goal: stop the few that grow large

Treemap: September and August drive almost all burned hectares

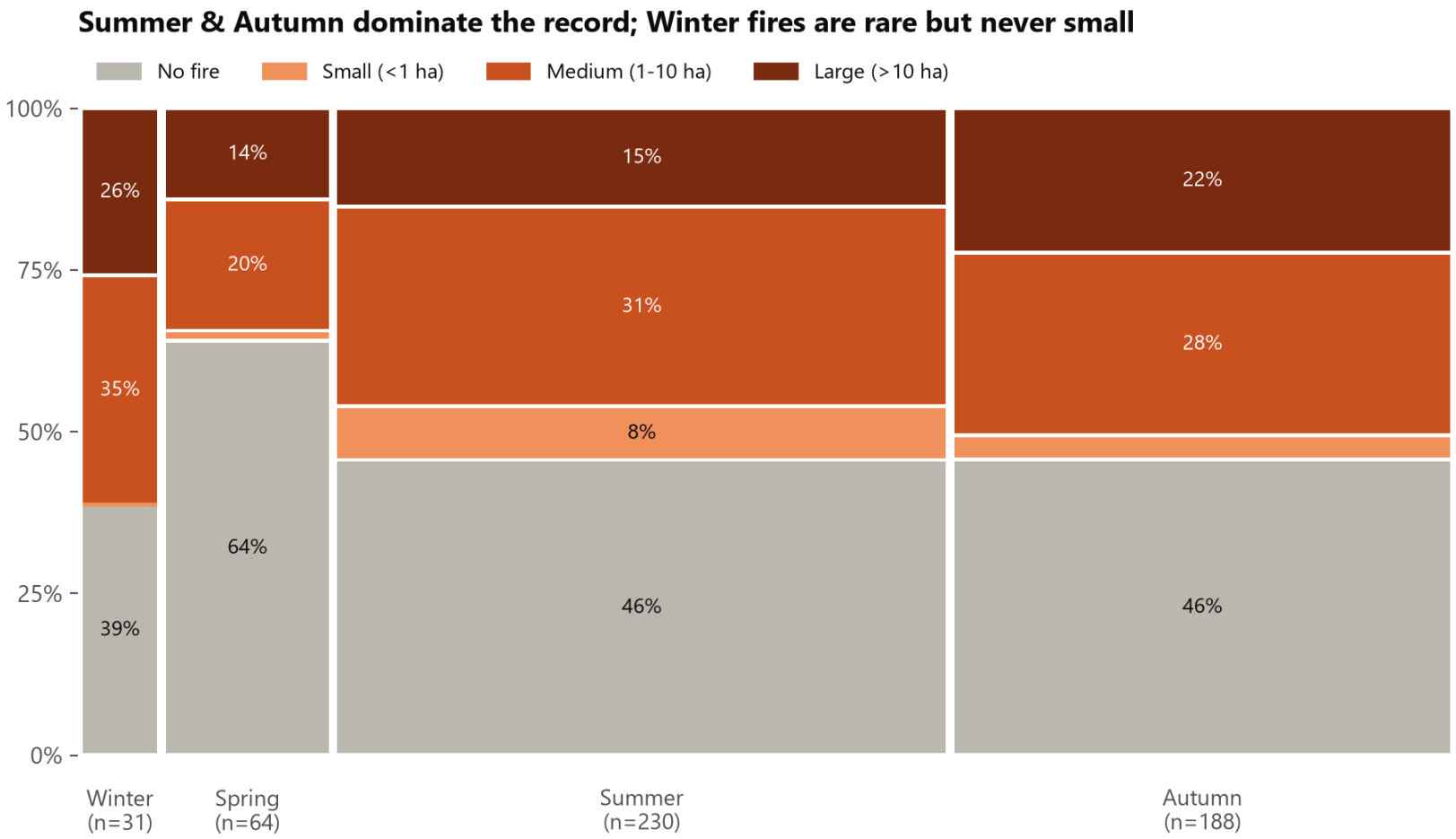
Where the hectares went: Summer and Autumn, driven by August and September



Reading the treemap

- **Size** = hectares burned
- **Colour** = share of observations with a fire
- Sep 3,086 ha · Aug 2,298 ha
- Spring & Winter are small slivers
- Dec is dark but n = 9 only

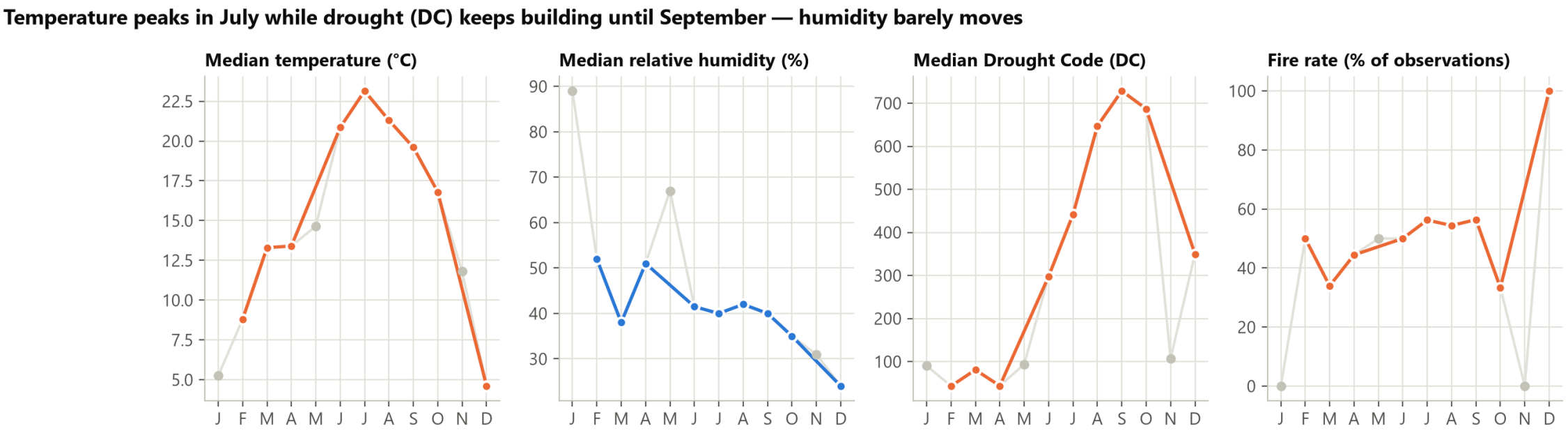
Mosaic plot: Autumn has the most large fires



Reading the mosaic

- **Width** = how many observations
- **Height** = mix of fire sizes
- Large fires: Autumn 22% vs Summer 15%
- Spring: 64% no-fire days
- Winter (n = 31): 0% small fires

Heat peaks in July, but drought keeps building until September



Temperature

Peaks in July at ≈ 23 °C

Humidity

Barely moves — around 40% all summer

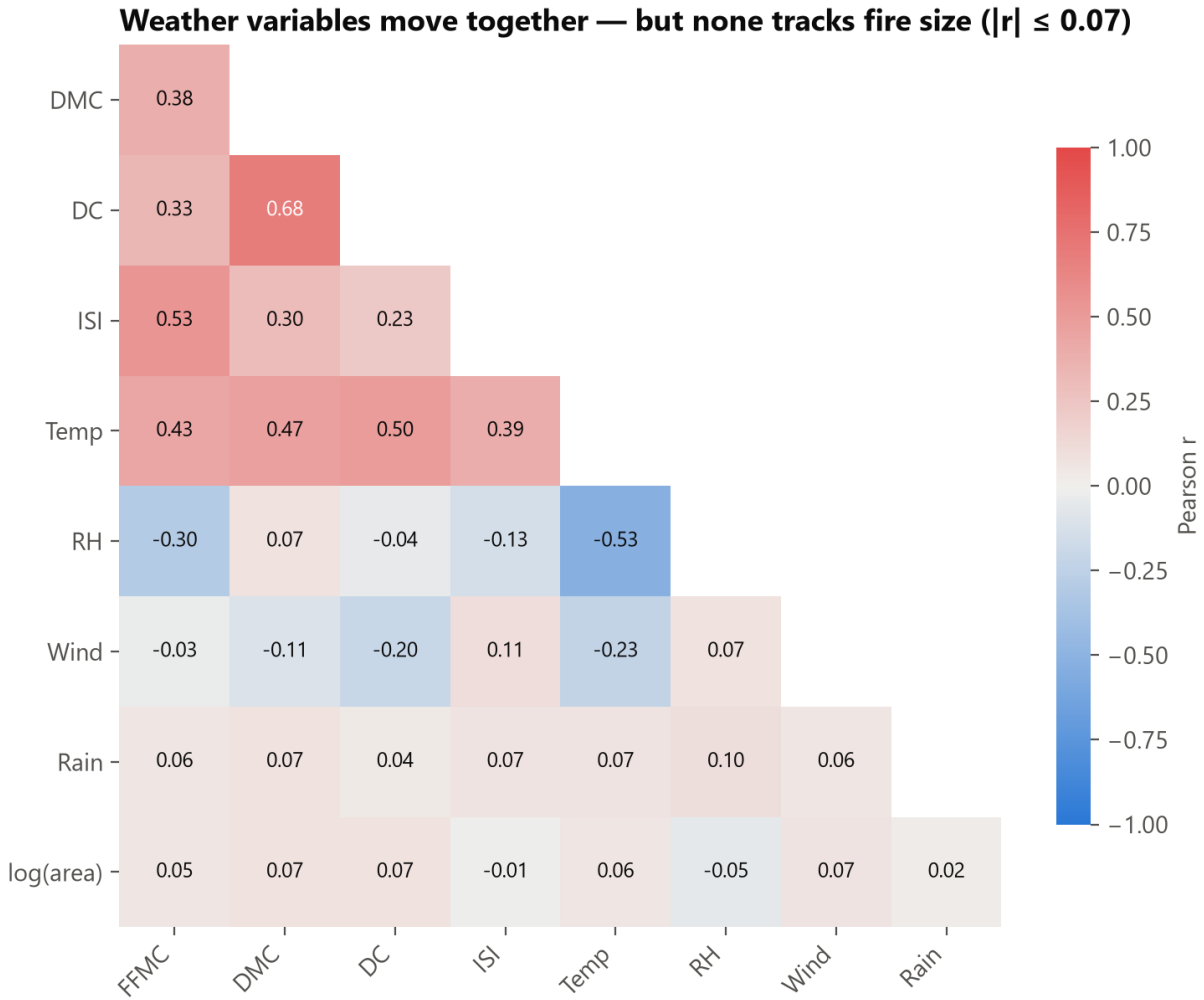
Drought Code

Rises to ≈ 730 in September

Fire rate

≈ 55% in July–September

Weather variables move together — but none tracks fire size



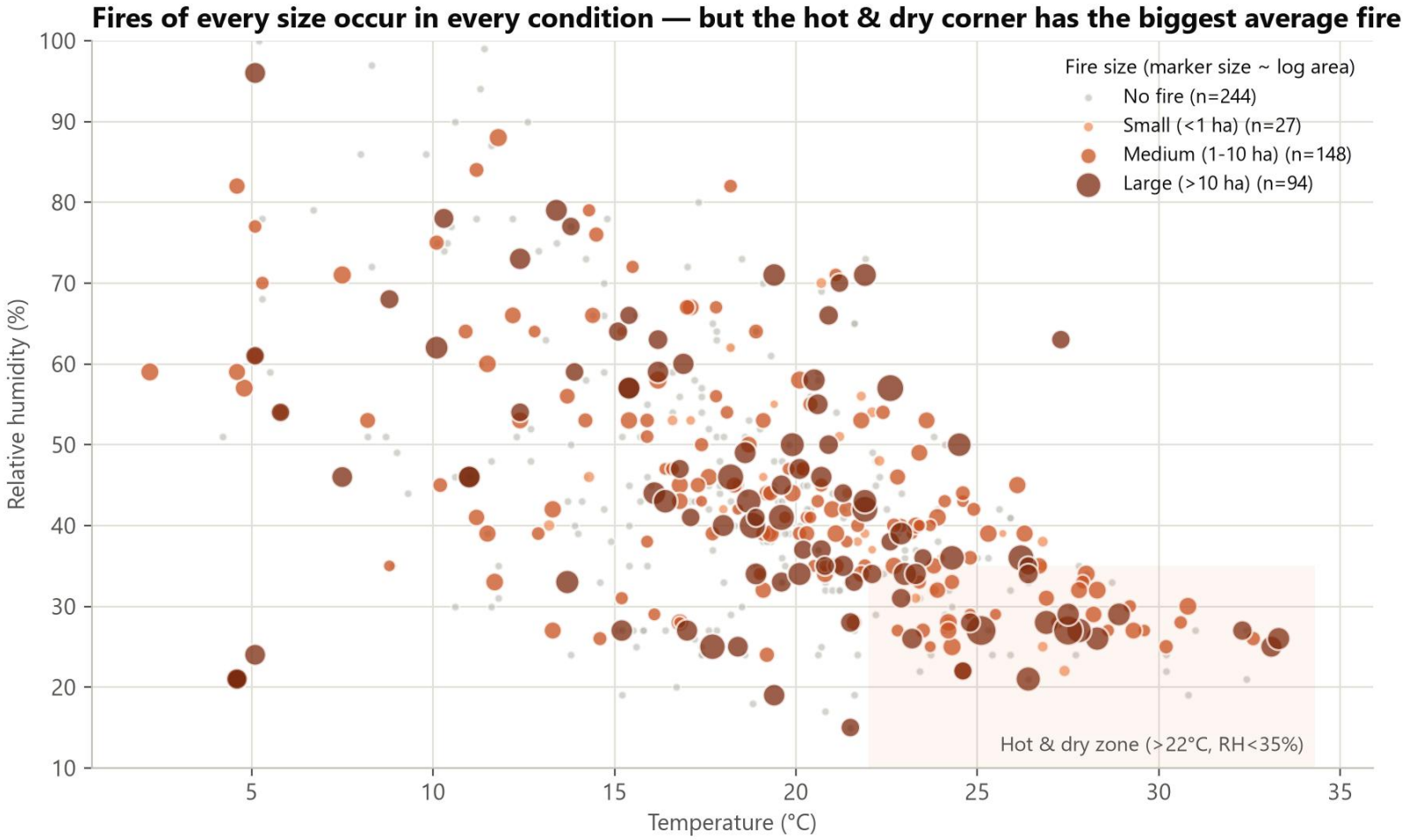
|r| ≤ 0.07

strongest correlation of any weather variable with fire size

What it means

- **DMC ↔ DC r = 0.68** — both measure accumulated dryness
- **Temp ↔ RH r = −0.53** — hot days are dry days
- **FFMC ↔ ISI r = 0.53**
- Weather sets the stage, but alone it does not decide how big a fire gets

Scatter: the hot & dry corner produces the biggest fires on average



Hot & dry vs other days

Average fire

30 ha vs 9 ha

Fire rate

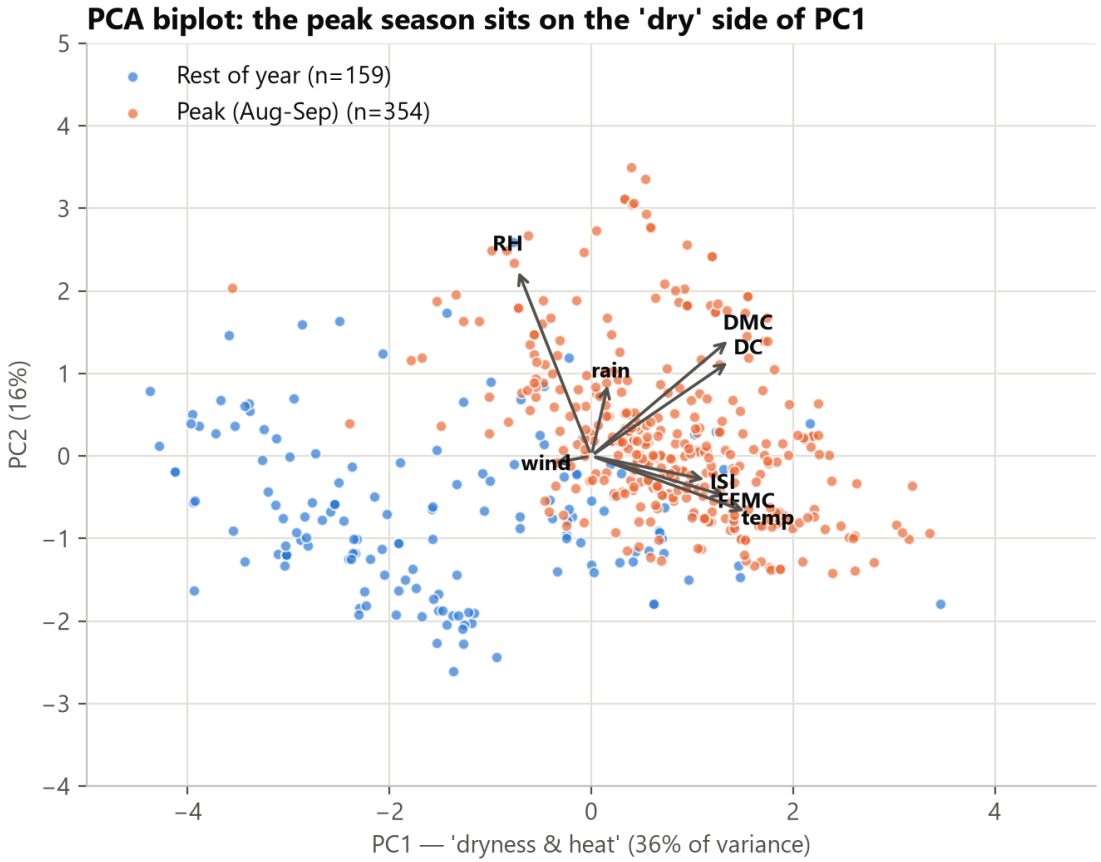
59% vs 51%

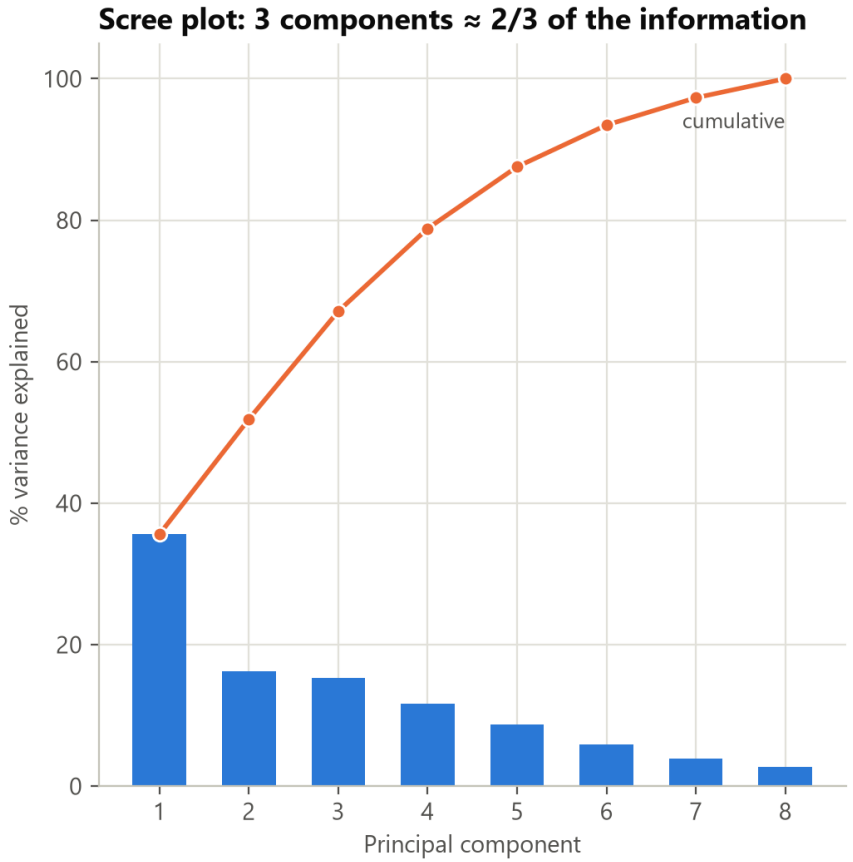
Large-fire rate

22% vs 18%

n = 87 hot & dry days

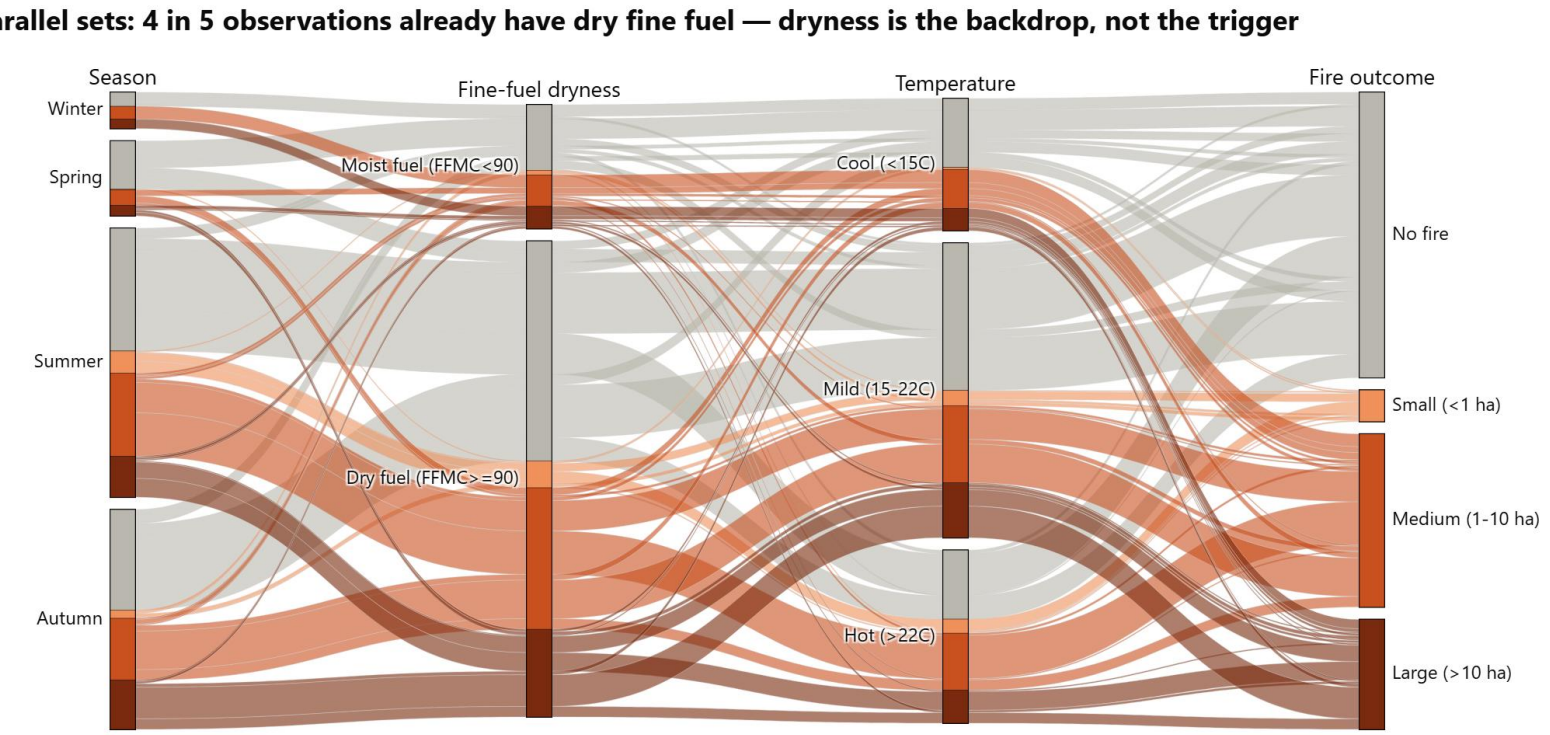
PCA: eight weather variables collapse onto one 'dryness & heat' axis





PC1 = **36%** of variance · peak-season days (orange) sit on the hot/dry side · 3 components keep **67%** of the information

Parallel sets: dry fuel is the backdrop of the season, not the trigger



79%

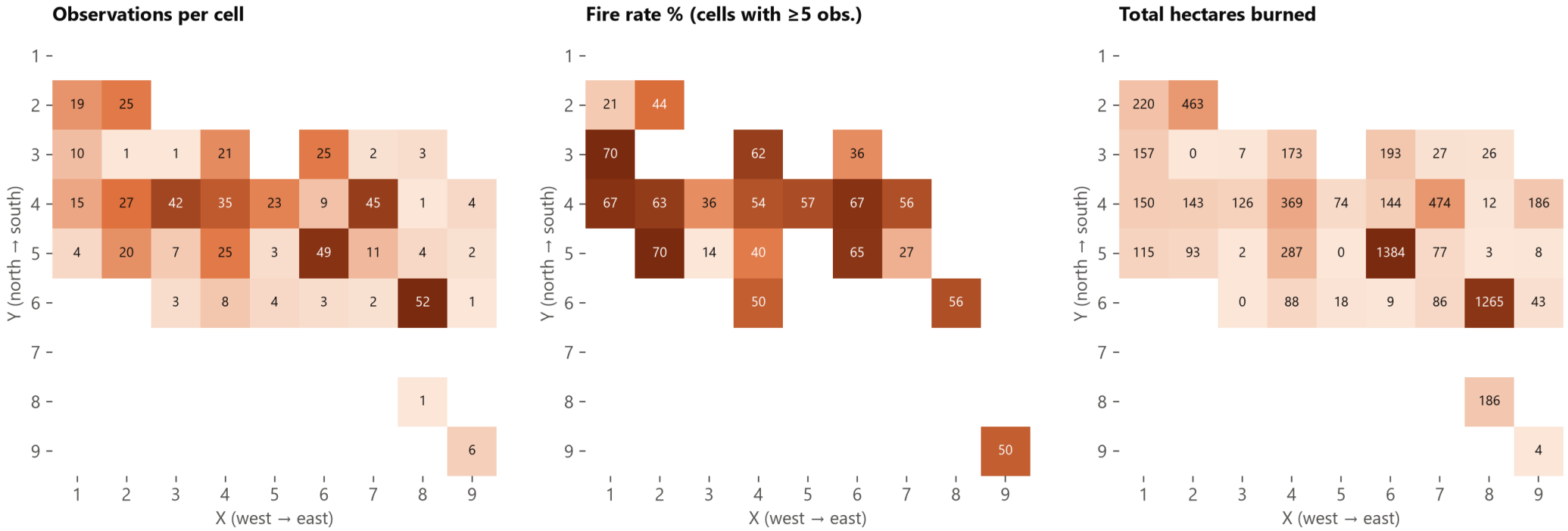
of observations already have dry fine fuel (FFMC ≥ 90)

What it means

- Dry-fuel ribbons reach every outcome
- An ignition is still needed — often human
- Combine weather with ignition data

Two grid cells hold 40% of all hectares burned

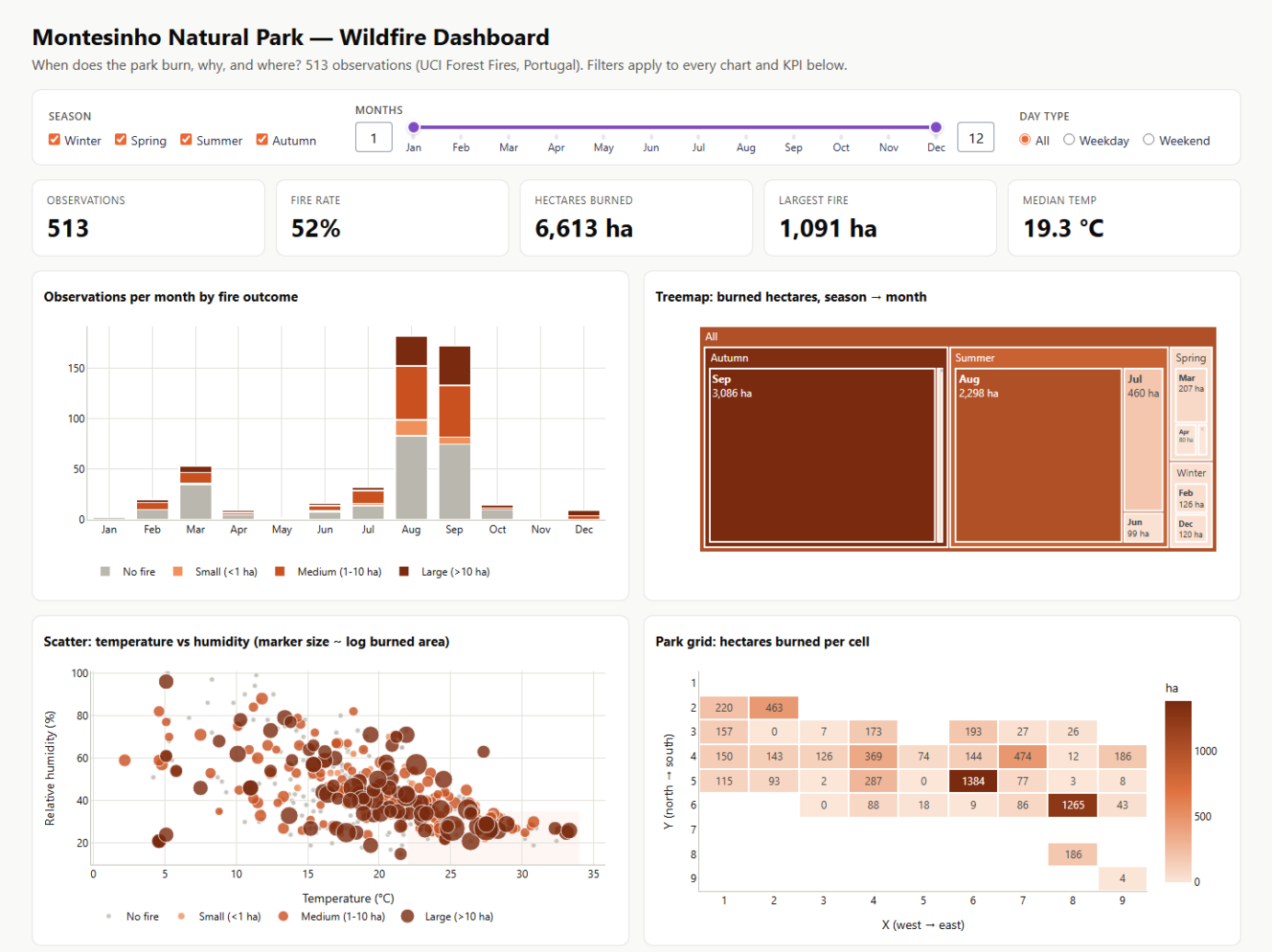
Spatial hotspots: two grid cells, (6,5) and (8,6), account for 40% of all hectares burned



Cell (X=6, Y=5) 1,384 ha burned · 32 fires in 49 observations

Cell (X=8, Y=6) 1,265 ha burned · 29 fires in 52 observations

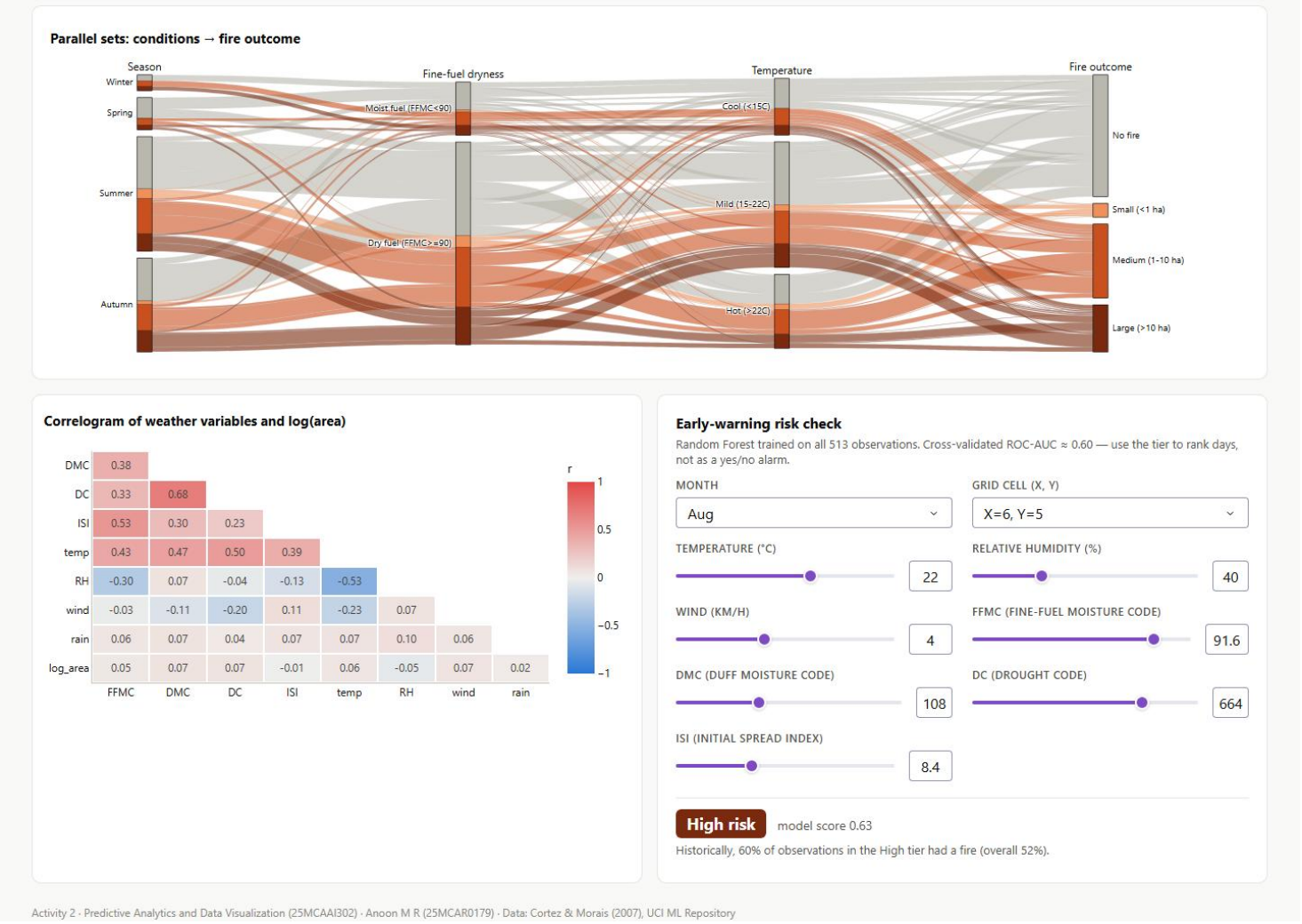
The dashboard: one filter row drives every chart and KPI



Built with Python

- **Filters:** season, months, weekday / weekend
- **KPIs:** observations, fire rate, hectares, largest fire, temperature
- **6 linked charts** with hover tooltips
- Offline copy: Wildfire_Dashboard.html

Early-warning risk check: from weather inputs to a risk tier

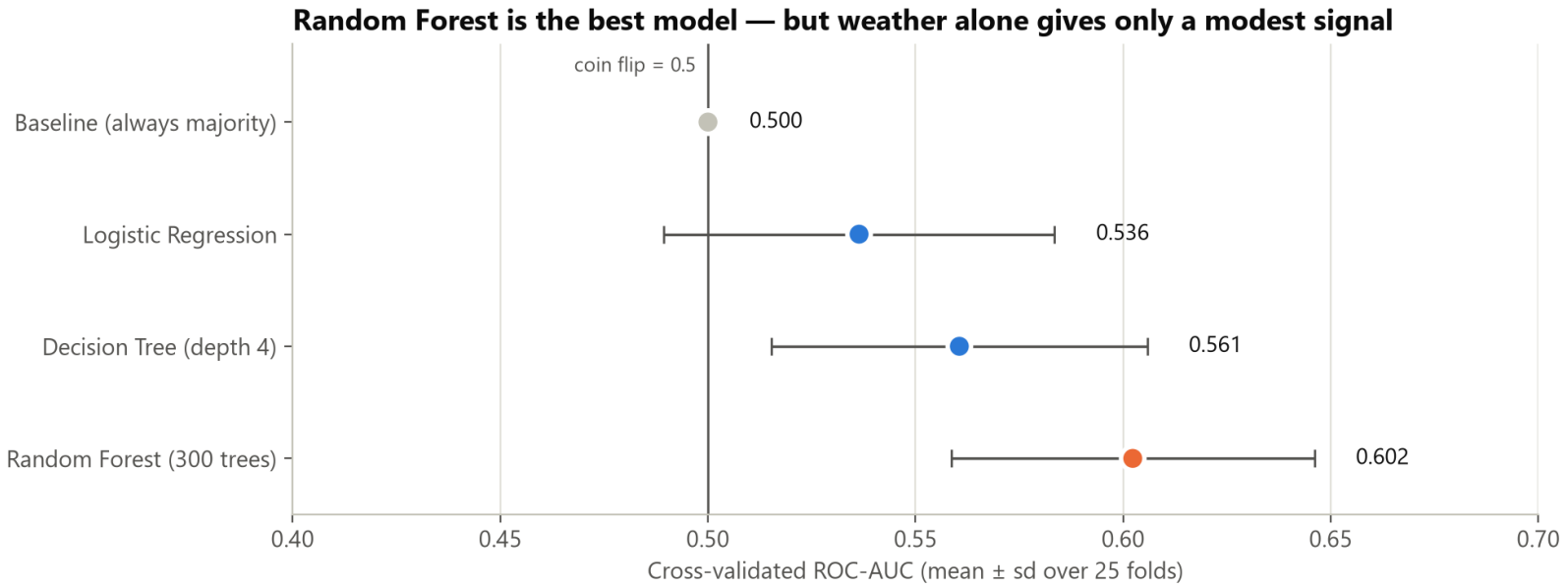


Activity 2 · Predictive Analytics and Data Visualization (25MCAAI302) · Anoon M R (25MCAR0179) · Data: Cortez & Morais (2007), UCI ML Repository

How officers use it

- Choose month, grid cell and the day's weather
- Random Forest returns **Low / Moderate / High**
- Shows the historical fire rate of that tier
- Used to **rank** days for patrols — not as a yes/no alarm

Random Forest wins — weather gives a modest but real signal



- **Target:** fire (any area burned) · 52% positive — balanced classes
- **Features:** location, month, day, FFMC, DMC, DC, ISI, temperature, humidity, wind, rain

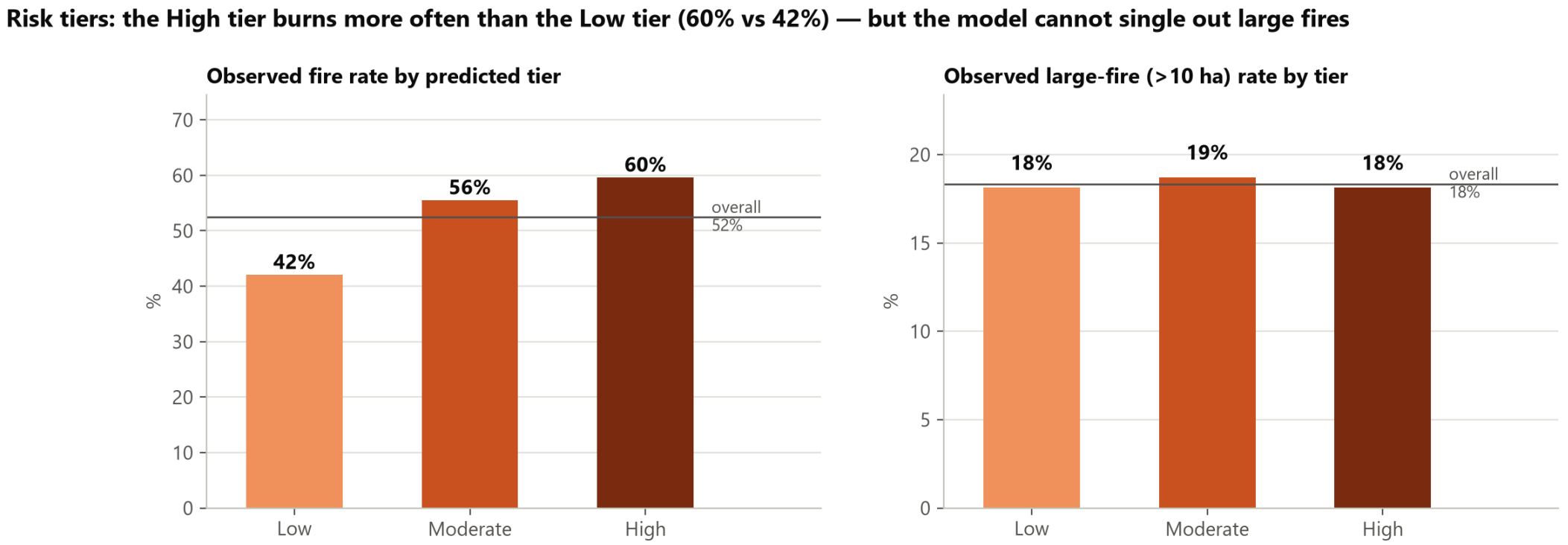
0.60

cross-validated ROC-AUC (± 0.04)

Honest evaluation

- 25 cross-validation fits per model
- Baseline = 0.50 (coin flip)
- Single split said 0.68 — too optimistic
- Top inputs: DMC, temperature, humidity

Risk tiers: High-tier days burned 60% of the time vs 42% for Low



Useful: the model ranks days by how likely a fire is. **Limit:** large fires are ≈ 18% in every tier — it cannot predict how big a fire will grow.

ACTIONABLE INSIGHTS

Six actions for the forest agency

1 Staff up for Aug–Sep

81% of hectares burn in these two months — concentrate crews and aerial patrols

2 Fast initial attack

The 10 largest fires = 51% of damage — stop fires before they grow

3 DC as readiness trigger

Raise readiness when the Drought Code passes ≈ 600 (typical of Aug–Sep)

4 Hot & dry day protocol

> 22 °C and RH < 35%: average fire 30 ha — pre-position tenders, restrict campfires

5 Protect two hotspots

Cells (6,5) and (8,6) = 40% of hectares — towers, cameras, fuel breaks

6 Rank days with the model

Use the dashboard risk tier to prioritise patrols — not as a yes/no alarm

CONCLUSION

Seasonal, concentrated, and driven by accumulated dryness

1

When

Aug–Sep = 81% of hectares

2

Why

Heat + drought build-up; weather sets the stage

3

Where

Two cells = 40% of hectares

4

What next

Risk tiers, CV ROC-AUC 0.60

Limitation: 513 observations from one park; weather explains when danger is high, not which exact plot will burn — ignition data (roads, tourism, lightning) is the next step.

Thank you

Questions?

Anoon M R · 25MCAR0179 · MCA AIML

Data: Cortez & Morais (2007), UCI Machine Learning Repository · Tools: Python, pandas, scikit-learn, Matplotlib, Plotly Dash